Master's Thesis

Reinforcement Learning-Based Feeder Link Switchover for Gateway Load Balancing in NTN

Minsik Kim

Department of AI Convergence Network

Graduate School

Ajou University

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by

Minsik Kim

Department of AI Convergence Network

Ajou University

Advisor : Professor Jae-Hyun Kim

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Abstract

This thesis proposes a deep reinforcement learning (DRL) framework for feeder link switchover (FLSO) to address gateway (GW) load balancing in low Earth orbit (LEO) non-terrestrial networks (NTN). Conventional FLSO strategies, which rely on simple distance or signal strength heuristics, often lead to severe load imbalance across ground GWs, degrading backhaul efficiency and network performance. To overcome this, this thesis formulates the FLSO decision problem as a Markov Decision Process (MDP) that jointly optimizes for link stability, signal quality, and GW load distribution.

This thesis introduces a novel FLSO algorithm based on a dueling double deep Q-network (DDQN) architecture, enhanced with prioritized experience replay (PER) and Polyak averaging. The agent learns to make intelligent switchover decisions using locally observable metrics, including satellite ephemeris, layer 3 (L3)-filtered RSRP, and real-time GW load indicators. The multi-objective reward function is designed to minimize unnecessary FLSOs (UFLSOs) and unstable links while maximizing connection duration (ToS), main-

taining high-quality RSRP, and actively balancing the load across GWs.

Through extensive simulations based on a realistic Starlink Gen2 constellation and United States-based GWs, this thesis demonstrates the superiority of proposed approach. Compared to 3rd generation partnership project (3GPP) inspired heuristic baselines which are signal-based and distance-based and a standard deep Q-network (DQN) agent-based, the proposed dueling DDQN-based policy achieves the best performance across key metrics. It simultaneously reduces the total number of FLSOs, eliminates UFLSOs, achieves the longest ToS, and, most critically, yields the most balanced GW load distribution. These results confirm that the proposed DRL-based agent effectively learns a sophisticated, load-aware FLSO policy that ensures stable, high quality backhaul connectivity without relying on manual heuristic tuning.

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List of Abbreviations

3GPP	3rd Generation Partnership Project
AMF	Access and Mobility Management Function
DDQN	Double Deep Q-Network
DNN	Deep Neural Network
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
ECEF	Earth-Centered, Earth-Fixed
EIRP	Effective Isotropic Radiated Power
FCC	Federal Communications Commission
FLSO	Feeder Link Switchover
FSPL	Free Space Path Loss
GEO	Geostationary Earth Orbit
GPU	Graphics Processing Unit
GW	Gateway
gNB	Next Generation NodeB
HO	Handover
IIR	Infinite Impulse Response

ISL	Inter-Satellite Link
KPI	Key Performance Indicator
L3	Layer 3
LEO	Low Earth Orbit
LoS	Line-of-Sight
MDP	Markov Decision Process
mMTC	massive Machine Type Communications
MTS	Minimum Time-of-Stay
NG	Next Generation interface
NLoS	Non-Line-of-Sight
NP	Non-deterministic Polynomial time
NR	New Radio
NTN	Non-Terrestrial Networks
PER	Prioritized Experience Replay
QoS	Quality of Service
RB	Resource Block
RL	Reinforcement Learning
RRM	Radio Resource Management
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SDN	Software-Defined Networking
TAC	Tracking Area Code
TD	Temporal-Difference
TLE	Two-Line Element set
ToS	Time-of-Stay

UE User Equipment

UFLSO Unnecessary Feeder Link Switchover

Chapter 1. Introduction

1.1. Background and motivation

Low Earth orbit (LEO) satellite networks have gained significant attention in recent years due to their potential to provide global broadband coverage with low latency. Operating at altitudes between 160 km and 2,000 km above the Earth's surface, LEO satellites offer several advantages over geostationary Earth orbit (GEO) satellites, including reduced signal propagation delay and lower power requirements for communication [1].

However, LEO satellites traveling at approximately 7–8 km/s make it difficult to maintain stable feeder and service links. In transparent-type satellites where the next generation nodeB (gNB) is at the gateway (GW), the satellite performs frequent feeder link switchovers (FLSO)s while passing multiple GWs; when the gNB differs across GWs, each FLSO triggers a simultaneous user equipment (UE) handover (HO) [2]. Such large-scale HOs, especially under 6G massive machine type communications (mMTC), cause signaling bursts that lead to HO failures and degraded service continuity [3]. For regenerative-type satellites, an

on-board gNB allows the UE to maintain its connection directly with the satellite even during a FLSO, meaning a UE HO does not occur as a direct result. However, these satellites still rely entirely on the feeder link for all backhaul traffic. Thus, while they avoid the massive UE HO problem associated with transparent types, the FLSO process itself selecting the optimal GW for backhaul remains a critical and frequent operational requirement for maintaining network connectivity. Moreover, heuristic FLSO strategies that prioritize distance-based or signal-based algorithm ignore the processing load and traffic distribution across ground GWs. As a consequence, specific GWs become severely congested while others remain underutilized, amplifying latency, wasting radio resources, and reducing the overall efficiency of the non-terrestrial network (NTN) backhaul.

1.2. Contributions

This thesis proposes a reinforcement learning (RL)-based FLSO scheme that optimizes the trade-off between link quality and GW load balancing. This thesis created a dynamic system environment to facilitate the RL agent's training process. The agent leverages locally observable metrics such as reference signal received power (RSRP), LEO satellite ephemeris data, and real-time GW load status. By incorporating these metrics, the agent adapts to rapid orbital dynamics and dynamic traffic, thereby enhancing link stability and reducing network congestion. This thesis designed a multi-objective reward function leveraging RSRP, time-of-stay

(ToS), and GW load to optimize long-term connectivity and network efficiency. This objective function is designed to manage the trade-offs between signal quality, link stability, and load balancing to minimize. This thesis proposed a dueling double deep Q-network (DDQN)-based FLSO algorithm to enhance robustness in noisy and dynamic LEO environments. This algorithm dynamically selects ground GWs to effectively balance network load while simultaneously reducing FLSO frequency, enhancing average RSRP, and extending ToS.

1.3. Overview

The remainder of this thesis is structured as follows: Section. II discusses LEO mobility, FLSO, and related research in load balancing, and RL in LEO NTN. Section. III presents the proposed RL framework in detail. Section. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Section. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite FLSO schemes

In NTN, LEO satellites are utilized to provide wide-area coverage while maintaining low end-to-end latency. Since LEO satellites move continuously along their orbits, global service is achieved through multi-satellite, multi-orbit constellations. Mobility management for LEO satellite networks is actively being standardized within 3rd generation partnership project (3GPP). Traditionally, 3GPP specifications focused on UE HO procedures between geographically fixed gNBs, where the next generation (NG) interface toward the core network remained essentially static and did not require dedicated mobility handling. However with LEO satellites, the satellite GW association itself becomes time varying, and the NG interface must track the movement of the satellite. As a result, 3GPP has begun to define explicit NG interface mobility and FLSO procedures for NTN, extending beyond conventional UE HO mechanisms to support this new form of dynamic connectivity [4].

In this context, the satellite payload architecture directly determines how this new mobility management translates into UE HO. For trans-

parent satellites, the gNB is located at the ground GW and UEs are served via the service link. When an FLSO changes the serving GW and the associated gNB, this backhaul level mobility is exposed to the access side, and a corresponding UE HO is triggered.

In contrast, a regenerative satellite has an on-board gNB. In this case, the UE remains anchored to the same gNB even if the feeder link is switched between different GWs, so an FLSO does not inherently require a UE HO. Instead, the impact of mobility is confined to the backhaul domain. Two practical FLSO execution modes are considered: hard FLSO, where the existing feeder link is released before the new one is established, and soft FLSO, where the new feeder link is established prior to releasing the old one.

Moreover, the FLSO process itself introduces significant procedural challenges, which are actively being addressed in 3GPP. As identified in [2], the hard FLSO mechanism can lead to uplink packet loss, as illustrated in Fig. 2.1. This occurs because the access and mobility management function (AMF), unaware of the on-board gNB's instantaneous switch, may continue to forward packets to the old GW after the satellite has already switched to the new GW. These vulnerabilities indicate that not only the execution of FLSO, but also the conditions under which it is triggered, must be carefully controlled to avoid unnecessary service disruption.

In NTN, this control is realized through standardized measurement events that determine when a HO should be initiated [5]. Multiple measurement event conditions are defined to trigger UE HO. For signal-based

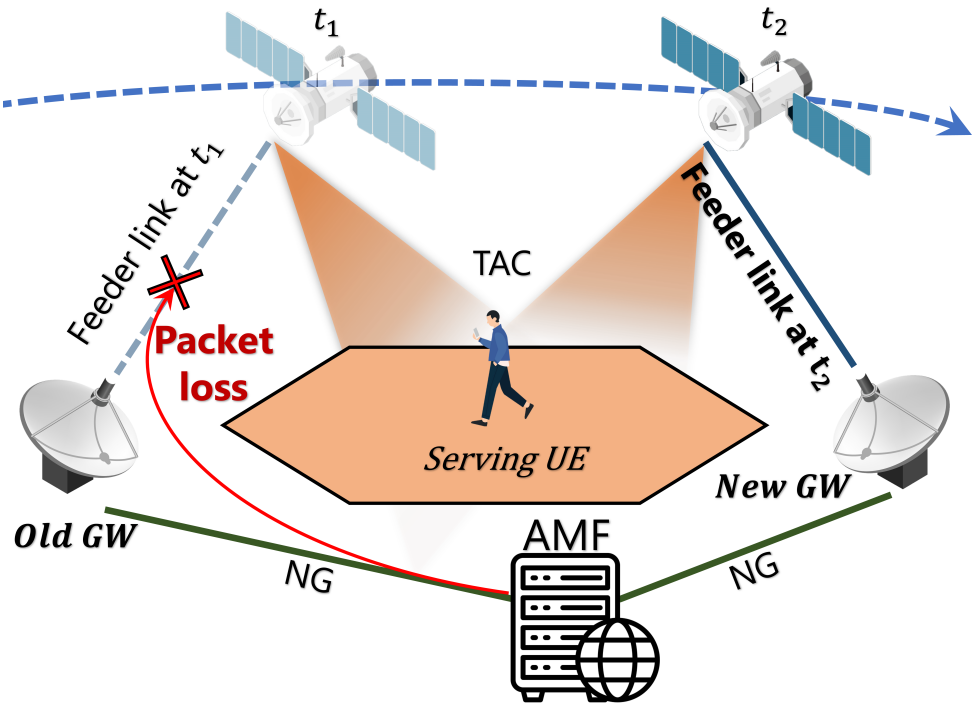


Fig. 2.1. Packet loss during FLSO process

triggers, as summarized in Table 2.1, the serving and target gNBs' RSRPs drive the HO decision; among them, Event A3 is most commonly used. However, for satellites operating at high altitude and moving rapidly, such signal-based events can be unstable. Moreover, on the ground the received signal levels often differ only marginally within a cell radius, making pure signal-based triggering ill-suited to NTN.

Therefore, 3GPP defines location-based measurement events that rely on absolute or moving reference locations, such as cell centers or ephemeris-derived beam centers, together with dual thresholds and hysteresis to mitigate ping-pong behavior [5]. Among these, the dual-threshold Events

Table 2.1. Entering conditions for measurement events

Event	Entering Condition
A1	Serving gNB's signal becomes stronger than the threshold.
A2	Serving gNB's signal becomes weaker than the threshold.
A3	Target gNB's signal becomes stronger than the serving gNB's signal by an Offset margin. (most common HO trigger)
A4	Target gNB's signal becomes stronger than the threshold.
A5	Serving gNB's signal becomes weaker than threshold 1, and Target gNB's signal becomes stronger than threshold 2.

D1 and D2 are most relevant to mobility in LEO NTN scenarios.

Event D1 is specified for static reference locations, such as serving and target cell centers. The triggering condition is evaluated based on the locations from the UE to two fixed reference points, using separate entering and leaving inequalities with a hysteresis margin to suppress rapid toggling.

Event D2 preserves the same decision structure but extends it to moving reference locations determined from satellite ephemeris, for example, time-varying satellite beam centers. This allows the dual-threshold and hysteresis mechanism to operate on dynamically shifting footprints, making D2 directly applicable to LEO NTN, where coverage areas evolve continuously.

Let M_{l1} and M_{l2} denote the locations from the UE to the two reference locations, Thresh_1 and Thresh_2 the corresponding location thresholds, and Hys the hysteresis margin. The generic entering and leaving

conditions can be written as

$$\text{Entering}: \quad (M_{l1} - \text{Hys} > \text{Thresh}_1) \wedge (M_{l2} + \text{Hys} < \text{Thresh}_2), \tag{2.1}$$

$$\text{Leaving}: \quad (M_{l1} + \text{Hys} < \text{Thresh}_1) \vee (M_{l2} - \text{Hys} > \text{Thresh}_2), \tag{2.2}$$

where all locations are measured in meters. Event D1 corresponds to the case in which the reference locations are static, whereas Event D2 applies the same dual-threshold and hysteresis form to ephemeris-based moving references. In both cases, the hysteresis shifts the entering condition outward and the leaving condition inward, ensuring that small fluctuations around the thresholds do not immediately retrigger the event.

In the LEO NTN context, D2-type location-based triggering is particularly attractive because the moving reference locations can track predicted satellite beam centers obtained from broadcast ephemeris information, enabling geometry-aware decisions under rapidly changing topology. However, these standardized location-based events remain essentially link-centric: they stabilize the trigger based on geometric and signal conditions at the UE side, but do not explicitly account for GW processing load or backhaul congestion. As a result, multiple satellites may converge on the same geometrically preferred GW, leading to feeder link bottlenecks. This motivates extending the location-based framework toward a GW-aware FLSO policy that jointly considers location, link stability, and GW load to improve overall NTN backhaul efficiency.

These procedural challenges underscore that FLSO is a complex and costly operation, not a simple link transition. Therefore, an optimal se-

lection policy must not only consider link quality but also minimize the frequency of unnecessary FLSOs.

While 3GPP defines the architectures for executing an FLSO, the optimal selection policy—determining which available GW to switch to—is non-standardized and remains a critical challenge for network performance.

2.2. Load Balancing in LEO Satellite Networks

Load balancing is a critical challenge in LEO satellite networks to ensure overall system efficiency and quality of service (QoS). The time-varying topology of the LEO constellation and the uneven global traffic demand can cause severe congestion on specific satellite nodes or links.

To solve this, much of the existing research has been concentrated on load balancing from a routing perspective via inter-satellite links (ISLs) [6–8]. These studies aim to manage the horizontal traffic flow between satellites to avoid congestion on the space-based backbone. These ISL-centric approaches include techniques such as SDN-based adaptive routing [6, 8] and back-pressure routing [7]. For instance, Roth et al. [6] propose a distributed software-defined networking (SDN)-based routing scheme that divides the constellation into clusters and performs load balancing for ISL paths within the cluster. Similarly, Deng et al. [7] propose a Distance-Based Back-Pressure Routing strategy, which calculates ISL link weights based on both the queue backlogs of satellite nodes and the propagation delay, thereby balancing traffic loads among satellite nodes.

Another domain of research focuses on radio resource management (RRM) at the service link level. These studies aim to balance the load among beams by efficiently allocating limited radio resources, such as beams, spectrum, or time slots, to UEs on the ground.

However, existing load balancing research, such as the ISL-centric routing schemes discussed above [6–8], is predominantly focused on balancing load between satellite ISLs or at the service link RRM level. In contrast, load balancing at the feeder link level, where the satellite constellation as a whole connects to the ground network remains a relatively unexplored area. Even if congestion at the satellite or ISL level is avoided, the terrestrial network bottleneck persists if all traffic ultimately converges on a single, overloaded GW.

As reviewed in Section 2.1, conventional FLSO strategies tend to rely solely on link-level metrics like RSRP or location. Because these methods do not consider the real-time processing load or traffic status of the GWs, they fail to perform intelligent load balancing. This paper addresses this gap by integrating GW load awareness into the FLSO decision-making process to optimize the overall efficiency of the NTN backhaul.

2.3. RL for LEO Mobility Management

The agent learns control policies through interaction with an environment: at each decision step the agent observes a state, selects an action, receives a scalar reward, and transitions to the next state [9]. The objective is to learn a policy that maximizes the discounted long-term return.

Value-based approaches approximate the action–value function $Q(s, a)$. A deep Q-network (DQN) predicts $Q(s, a)$ and selects the action with the highest predicted value. To improve stability and accuracy, common enhancements include DDQN, which decouples action selection and evaluation, dueling architecture, which separates value and advantage streams, prioritized replay, which samples transitions with large temporal difference (TD) errors more often, and Polyak averaging for smooth target updates. Function approximators are kept lightweight yet expressive—typically multi-layer perceptrons with normalization and nonlinear activations—and feasibility is enforced by action masking so that invalid actions are eliminated before scoring. Parameters are updated via backpropagation using Adam or AdamW. Exploration in value-based agents follows an ϵ-greedy schedule, whereas policy-based methods rely on entropy regularization that is gradually annealed. Experience replay breaks temporal correlations and increases sample reuse, and the Huber (smooth-L_1) loss is preferred for TD regression due to its robustness to outliers and nonstationary targets.

Recent studies [10, 11] have applied deep reinforcement learning (DRL) to NTN mobility subproblems, including service link HO and beam resource scheduling, reporting improved throughput and reduced delay compared with heuristic baselines. However, most designs assume centralized observability or ignore the coupling to the feeder link, and thus do not address GW selection or backhaul load balancing explicitly.

Chapter 3. Proposed RL-based FLSO algorithm

3.1. System model

This thesis considers a large-scale LEO satellite constellation deployed across P orbital planes, represented by the set $\mathcal{P} = \{p \mid p = 1, \ldots, P\}$. The constellation consists of N LEO satellites in total, distributed across these planes, with N_p satellites in plane p. The overall satellite index set is denoted as $\mathcal{S} = \{s_{p,i} \mid p = 1, \ldots, P;\ i = 1, \ldots, N_p\}$, and its cardinality as $|\mathcal{S}| = \sum_{p=1}^{P} N_p$. For simplicity in the following equations, a single satellite is denoted by its global index $n \in \{1, \ldots, N\}$. The LEO satellite n connects to ground GWs via feeder link, where the set of GWs is denoted as $\mathcal{G} = \{m \mid m = 1, 2, \ldots, M\}$. The coordinates of each LEO satellite n in the Earth-centered, Earth-fixed (ECEF) coordinate system at a given time slot t, where $t \in \{0, 1, \ldots, T\}$, are given by $l_n(t) = (x_n(t), y_n(t), z_n(t))$.

The primary challenge addressed in this thesis is GW load balancing. The load of a GW m at time t, denoted as $\mathcal{L}_m(t)$, is defined as the set of satellites $n \in \mathcal{S}$ currently connected to it. The instantaneous load is

$|\mathcal{L}_m(t)|$.

The LEO satellite n evaluates the RSRP from signals transmitted by the GW m at periodic interval T_m. The RSRP $R_{m,n}$ is calculated as follows:

$$R_{m,n} = EIRP_m + G_n - PL_{m,n} - 10\log_{10}(q_{RB}), \tag{3.1}$$

where $EIRP_m$ is the effective isotropic radiated power (EIRP) of the GW m, defined as $EIRP_m = P_m + G_m$, with P_m being the transmit power and G_m the transmit antenna gain of the GW m towards the LEO satellite n. G_n represents the receive antenna gain of the LEO satellite n. The term $10\log_{10}(q_{RB})$ is used to normalize the RSRP measurement, accounting for a single resource block out of the total resource blocks q_{RB} allocated to the link. $PL_{m,n}$ is the large-scale path loss between the GW m and the LEO satellite n, and this expression is adapted from 3GPP documentation [12]. The model is calculated as follows:

$$\begin{aligned} PL_{m,n} &= P_{LOS,m,n}\big(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n}\big) \\ &\quad + P_{NLOS,m,n}\big(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n} + CL_{m,n}\big) + AT_{m,n}, \end{aligned} \tag{3.2}$$

$P_{LOS,m,n}$ and $P_{NLOS,m,n}$ denote the probabilities that the LEO satellite n to GW m is in a line-of-sight (LoS) and non-line-of-sight (NLoS) condition, respectively. $FSPL_{m,n}(d_{m,n}, f_m)$ denotes the free space path loss (FSPL) as a function of the distance $d_{m,n}$ between the LEO satellite n and the GW m and the frequency f_m of the GW m uplink feeder link frequency. $SF_{m,n}$ represents the shadow fading, which accounts for the signal attenuation caused by obstacles such as buildings, trees, or terrain vari-

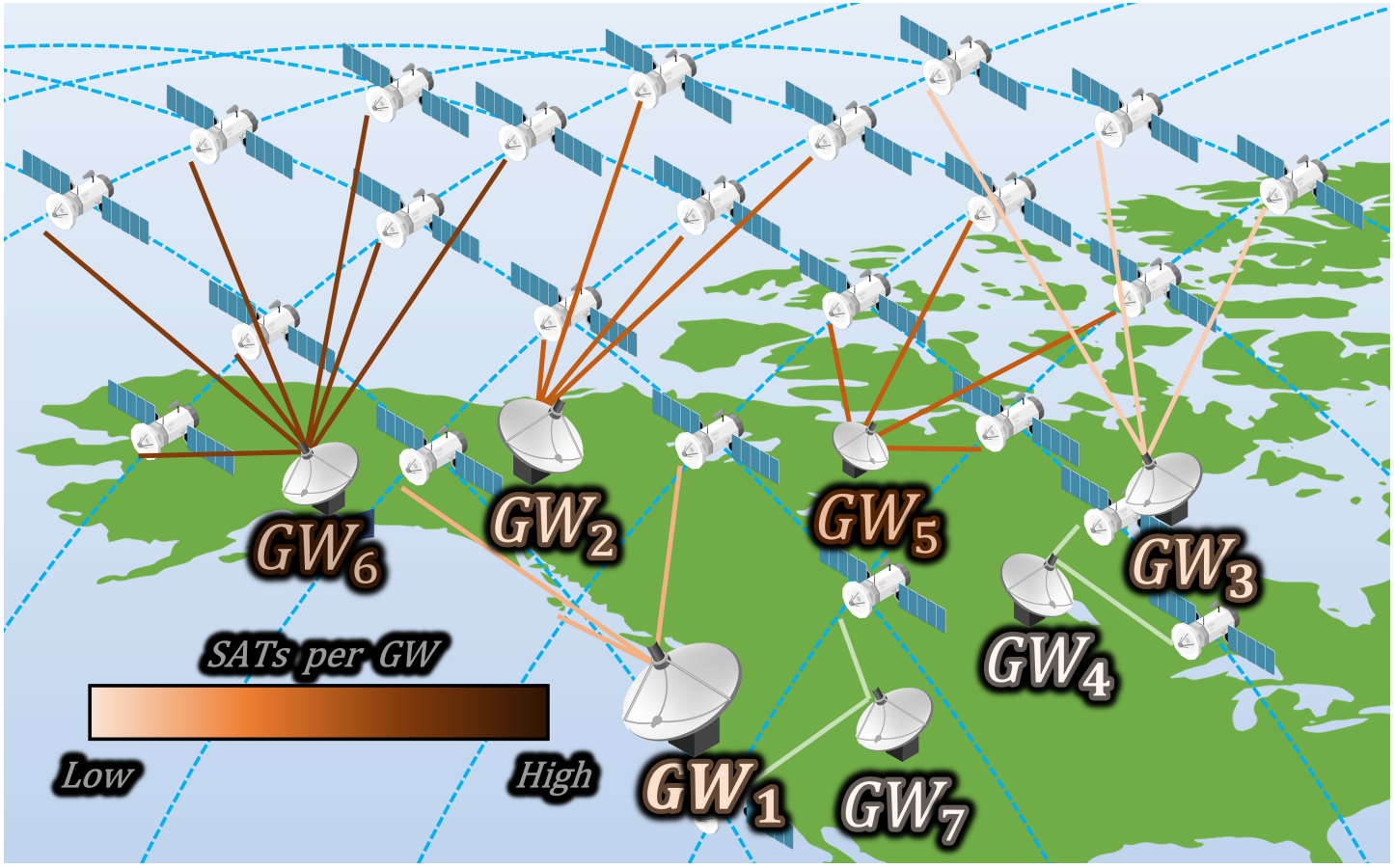


Fig. 3.1. Network scenario of a LEO satellite performing FLSO while across Starlink GWs in the United States.

ations in the propagation path. $CL_{m,n}$ denotes the clutter loss, reflecting additional signal degradation due to environmental clutter. $AT_{m,n}$ captures atmospheric attenuation along the feeder link slant path, including gaseous absorption due to dry air and water vapor, and it can be obtained from standard ITU-R models for the considered band [13]. Specifically, the $FSPL_{m,n}$ is calculated as follows:

$$FSPL_{m,n} = 32.45 + 20\log_{10}(f_m) + 20\log_{10}(d_{m,n}). \tag{3.3}$$

At each measurement instant t, the LEO satellite n samples the RSRP from GW m and all neighboring GWs within instantaneous visibility region, yielding the raw RSRP $R_{m,n}(t)$. To mitigate the effects of shadow fading and measurement noise, these raw RSRP $R_{m,n}(t)$ samples are processed at the radio resource control (RRC) layer using a layer 3 (L3) infinite impulse response (IIR) filter [14]. Denoting the filtered RSRP by $\widetilde{R}_{m,n}(t)$, the update equation is calculated as follows:

$$\widetilde{R}_{m,n}(t) = aR_{m,n}(t) + (1-a)\widetilde{R}_{m,n}(t-1), \tag{3.4}$$

where $a \in (0,1)$ is the L3 forgetting factor, defined as $a = \left(\frac{1}{2}\right)^{K/4}$, and K is the filter coefficient. The initial condition is set as $\widetilde{R}_{m,n}(0) = R_{m,n}(0)$. Finally, the LEO satellite compiles a measurement report containing the L3-filtered RSRP $\widetilde{R}_{m,n}(t)$ of the serving and neighboring GWs and transmits this report back to the serving GW m over the feeder link.

3.2. Problem formulation

This section addresses the optimization problem for the LEO satellite constellation. The objective is to find an optimal FLSO policy π that dynamically assigns each satellite $n \in \mathcal{S}$ to a GW $m \in \mathcal{G}$ at each time slot t.

The primary goal is to minimize the overall network load imbalance while simultaneously maximizing the collective link quality RSRP and minimizing the instability caused by frequent FLSOs. Let m_t^n denote the GW selected by satellite n at time t. This problem can be formulated as a multi-objective optimization problem.

The following cost and reward components are defined for each satellite n at time t:

- **FLSO Cost:** A cost C_{FS} is incurred if the satellite switches its serving GW, penalizing instability.

$$FS_{m,n}(t) = \begin{cases} C_{FS} & \text{if } m_t^n \neq m_{t-1}^n, \\ 0 & \text{otherwise.} \end{cases} \tag{3.5}$$

- **UFLSO Cost:** A cost C_{UFS} is incurred for an unnecessary FLSO (UFLSO), which occurs if the satellite switches to another GW within a minimum time-of-stay (MTS), denoted τ_{min}. Let $\tau_{stay}(t)$

is the time elapsed since the last FLSO.

$$UFS_{m,n}(t) = \begin{cases} C_{UFS} & \text{if } m_t^n \neq m_{t-1}^n \text{ and } \tau_{stay}(t) < \tau_{min}, \\ 0 & \text{otherwise.} \end{cases} \tag{3.6}$$

- **ToS Reward:** A reward is gained for maintaining a connection with the same GW, proportional to the duration of the connection.

$$ToS_{m,n}(t) = \tau_{stay}(t). \tag{3.7}$$

- **Load Imbalance Cost:** A cost $|\mathcal{L}_m(t)|$ is incurred for connecting to a congested GW, proportional to the number of satellites already connected.

- **Link Quality Reward:** A reward is gained based on the L3-filtered RSRP from the selected GW.

$$R_{m,n}(t) = \widetilde{R}_{m,n}(t). \tag{3.8}$$

The global optimization problem **P1** is to find the policy π that minimizes the total expected cumulative cost and maximizes rewards for all N satellites over the time horizon T.

$$\mathbf{P1:} \quad \min_{\pi} \mathbb{E} \sum_{t=0}^{T} \sum_{n=1}^{N} \sum_{m=1}^{M} \Big(FS_{m,n}(t) + UFS_{m,n}(t) - ToS_{m,n}(t) - R_{m,n}(t) + |\mathcal{L}_m(t)| \Big), \tag{3.9}$$

s.t.

$$\mathbf{C1:} \quad \theta_{m,n}(t) \geq \theta_{\min}, \quad \forall n, \forall m \in \mathcal{G}_t^n, \forall t, \tag{3.10}$$

$$\mathbf{C2:} \quad \widetilde{R}_{m,n}(t) \geq R_{\min}, \quad \forall n, \forall m \in \mathcal{G}_t^n, \forall t, \tag{3.11}$$

$$\mathbf{C3:} \quad m_t^n \in \mathcal{G}_t^n, \quad \forall n, \forall t. \tag{3.12}$$

In **P1**, the objective function jointly considers switching stability, connection duration, link quality, and GW load to achieve balanced and robust network operation.

The constraints define the set of available GWs, $\mathcal{G}_t^n$, for satellite n at time t. Constraint **C1** ensures the elevation angle $\theta_{m,n}(t)$ to an available GW m exceeds the minimum threshold $\theta_{\min}$, guaranteeing a reliable LoS. Constraint **C2** ensures the filtered RSRP $\widetilde{R}_{m,n}(t)$ from an available GW exceeds the minimum required signal quality $R_{\min}$. Constraint **C3** mandates that each satellite n must select a GW m_t^n from the set of currently available GWs $\mathcal{G}_t^n$ that satisfy C1 and C2.

This problem is characterized by high-dimensional state spaces and complex, time-varying dynamics, making it non-deterministic polynomial-time hard (NP-hard) [15]. This thesis therefore models the problem as a Markov decision process (MDP) to find an optimal FLSO policy using RL techniques.

3.3. Proposed method

To address the problem **P1**, the problem is transformed into a dynamic programming framework to derive an optimal FLSO policy. The FLSO optimization problem is modeled as an MDP due to its ability to handle sequential decision-making in dynamic environments.

3.3.1 MDP modeling

The LEO satellite's state s_t evolves over time and actions a_t impact the future reward r_t with discount factor γ [16]. This MDP formulation enables the agent to jointly account for geometric visibility, link quality, and GW load. At each step t, the state of satellite n is defined as

$$s_t = \left\{ l_n(t),\, m_t,\, \widetilde{R}_{m,n}(t),\, c_m(t) \right\}, \tag{3.13}$$

where $l_n(t)$ denotes the ECEF position of satellite n, m_t is the index of the serving GW selected at the current step, $\widetilde{R}_{m,n}(t)$ is the L3-filtered RSRP from that serving GW, and $c_m(t) \in [0, 1]$ is the normalized backhaul complexity of the serving GW. The complexity term $c_m(t)$ is obtained by mapping the instantaneous or pre-estimated number of attached satellites at GW m to a dimensionless score in $[0, 1]$ via min-max normalization across all GWs, so that higher values indicate more congested GWs. This compact representation provides the DRL agent with a load-aware yet scale-invariant signal, allowing it to avoid persistently overloaded GWs while optimizing FLSO decisions.

The action space a_t involves the LEO satellite n selecting the optimal GW $m \in \mathcal{G}$ at each step. In line with constraints **C1-C3**, the action space is the set of available GWs that meet the minimum elevation and RSRP thresholds:

$$a_t = \{m_t | m_t \in \mathcal{G}_t^n\}. \tag{3.14}$$

The action a_t is the selection of the next GW, m_t, from the set of available GWs $\mathcal{G}_t^n$.

The immediate reward r_t is formulated to directly optimize the objective function defined in **P1**. It penalizes FLSO costs and high load connections while rewarding high quality RSRP links and connection stability. It is calculated as follows:

$$r_t = w_R R_{m,n}(t) - w_S \Big(FS_{m,n}(t) + UFS_{m,n}(t) \Big) - w_T ToS_{m,n}(t) - w_L c_m(t), \tag{3.15}$$

where the cost and reward components are defined in Eq. (3.5) through (3.8), and the weights are the corresponding weights from **P1**.

Discount factor γ is a value between 0 and 1, which represents the rate at which future rewards are discounted when considering them at the current step.

3.3.2 Proposed dueling DDQN algorithm

This thesis employs a RL algorithm to enable LEO satellites to operate in an dynamic environment. As depicted in Fig.3.2, the architecture of the proposed dueling DDQN algorithm, incorporating a ϵ-greedy policy, is designed to optimize FLSO in LEO satellite networks. The LEO satellite receives feedback to execute actions that sustain both a prolonged ToS and an elevated RSRP, while minimizing the frequency of FLSO and, critically, GW congestion. This thesis enhancs the traditional DQN framework by incorporating DDQN, prioritized experience replay (PER), and dueling DQN to improve stability, learning efficiency, and robustness against noise in LEO satellite communication systems.

The DQN algorithm [17] approximates the Q-function $Q(s_t, a_t; \theta)$, which maps a state s_t and action a_t to an expected cumulative reward, using a deep neural network (DNN) parameterized by θ. Two DNNs are utilized: an online network for action selection and a target network for value estimation. The Q-function is trained to minimize the following loss function and loss function is calculated as follows:

$$L(\theta) = \mathbb{E}\left[\left(y_t^{DDQN} - Q(s_t, a_t; \theta)\right)^2\right], \tag{3.16}$$

where the y_t^{DDQN} is the target and calculated as follows:

$$y_t^{DDQN} = r_t + \gamma \max_a Q(s_{t+1}, \arg\max_{a'} Q(s_{t+1}, a'; \theta); \theta^-), \tag{3.17}$$

where θ^- represents the parameter of the target network. The target

network parameter are updated as $\theta^- \leftarrow \tau \cdot \theta + (1-\tau) \cdot \theta^-$, ensuring slow and stable updates to mitigate the moving target problem in the dynamic LEO satellite environment [18].

To improve robustness against random noise, such as channel shadow fading, this thesis utilizes a dueling DQN [9]. The dueling DQN architecture enhances the traditional DQN by splitting the Q-function into two separate streams: the state-value function $V(s;\theta)$ and the advantage function $A(s,a;\theta)$. The Q-function is then calculated as follows:

$$Q(s,a;\theta) = V(s;\theta) + \left(A(s,a;\theta) - \frac{1}{|A|} \sum_{a'} A(s,a';\theta) \right), \tag{3.18}$$

where $V(s;\theta)$ represents the value of the state s, $A(s,a;\theta)$ represents the advantage of taking action a in state s, and $|A|$ is the number of possible actions. This decomposition allows the network to better distinguish the value of a state from the advantage of each action, improving robustness against noise and enhancing learning efficiency in complex LEO satellite environments.

To further enhance the learning efficiency, this thesis incorporates PER as proposed in [19]. PER prioritizes experiences with higher TD errors, where the TD error δ_i for transition i is defined as the difference between the target and current Q-values. The priority is calculated as follows:

$$p_i = |\delta_i| + \epsilon_p, \tag{3.19}$$

where $\delta_i = r_i + \gamma Q(s_{i+1}, a_{i+1}; \theta^-) - Q(s_i, a_i; \theta)$ is the TD error, and

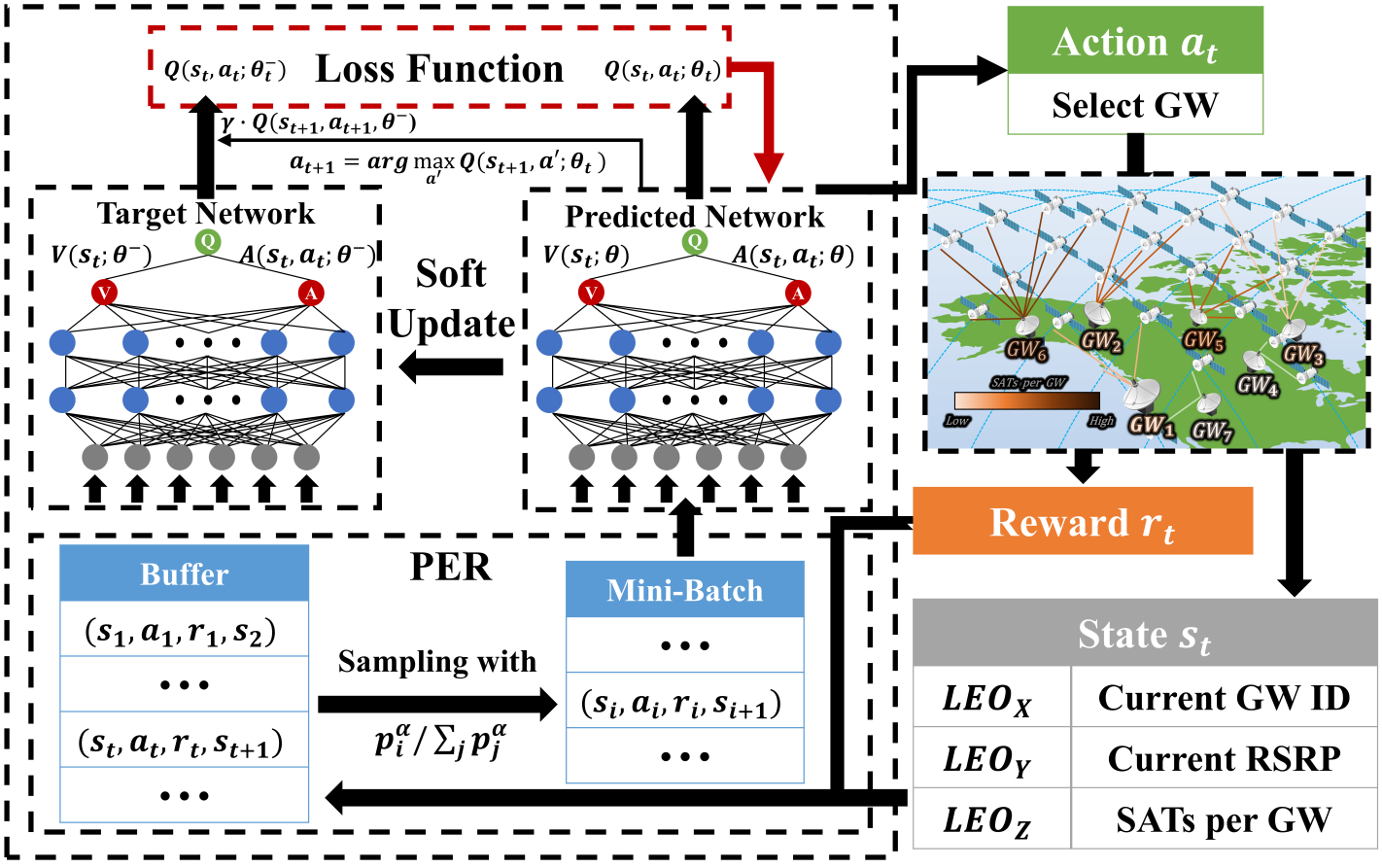


Fig. 3.2. The architecture of the proposed dueling DDQN for FLSO optimization in LEO satellite networks.

ϵ_p ensures non-zero priorities. PER uses $\alpha \in [0, 1]$ to control sampling probability p_i^α and $\beta \in [0, 1]$ to adjust importance sampling weights $(\mathcal{D} \cdot P(i))^{-\beta}$, where $\mathcal{D}$ is replay buffer limit and $P(i)$ is the probability of sampling transition i. This approach accelerates learning by focusing on significant transitions, which is beneficial in LEO satellite networks with dynamic and noisy conditions.

By combining dueling DQN with DDQN and PER, this approach effectively addresses overestimation bias, accelerates learning, and mitigates the effects of random noise, leading to improved network performance with effective GW load balancing, elevated RSRP, and reduced FLSO frequency. The detailed process of the training is described in Algorithm **??**.

Algorithm 3.1 Dueling DDQN-based FLSO Load Balancing

Require: Replay buffer capacity $\mathcal{D}$, batch size $\mathcal{Z}$, episodes $\mathcal{E}$, discount γ, PER parameters $(\alpha, \beta, \epsilon_p)$, exploration ϵ, Polyak coefficient τ, learning rate η

Ensure: Optimal FLSO policy π (GW selection)

Initialize: online network parameters θ (random), target network $\theta^- \leftarrow \theta$, replay buffer $\mathcal{R}$ with capacity $\mathcal{D}$
for *Episode* $= 1$ to $\mathcal{E}$ **do**
    Reset integrated LEO–GW environment
    for $t = 0$ to T **do**
        Observe state s_t
        Action selection (ϵ-greedy):
        $a_t = \begin{cases} \arg\max_{a'} Q(s_t, a'; \theta) & \text{with } 1 - \epsilon \\ \text{random action} & \text{with } \epsilon \end{cases}$
        Execute a_t, get reward r_t and next state s_{t+1}
        Store transition (s_t, a_t, r_t, s_{t+1}) into $\mathcal{R}$
        if $|\mathcal{R}| \geq \mathcal{Z}$ **then**
            Sample minibatch using PER:
            for $i = 1$ to $\mathcal{Z}$ **do**
                Sample $(s_i, a_i, r_i, s_{i+1}) \sim \mathcal{R}$ with $P(i) = \dfrac{p_i^\alpha}{\sum_k p_k^\alpha}$
                DDQN target:
                $y_i \leftarrow r_i + \gamma\, Q(s_{i+1}, \arg\max_{a'} Q(s_{i+1}, a'; \theta); \theta^-)$
                TD error and priority update:
                $\delta_i \leftarrow y_i - Q(s_i, a_i; \theta)$, $p_i \leftarrow |\delta_i| + \epsilon_p$
                Importance sampling weight:
                $w_i \leftarrow (\mathcal{D} \cdot P(i))^{-\beta}$
            end for
            Normalize $\{w_i\}$ by $w_i \leftarrow w_i / \max_j w_j$
            Weighted Huber loss:
            $L \leftarrow \dfrac{1}{\mathcal{Z}} \sum_{i=1}^{\mathcal{Z}} w_i\, \mathcal{L}_{\text{Huber}}(\delta_i)$
            Update online network:
            $\theta \leftarrow \theta - \eta \nabla_\theta L$
            Polyak update of target network:
            $\theta^- \leftarrow \tau\, \theta + (1 - \tau)\, \theta^-$
        end if
    end for
end for

Chapter 4. Performance evaluation

4.1. Simulation setup

In this section, the performance of the proposed algorithm is evaluated alongside DQN-based, distance-based, and signal-based FLSO algorithms through extensive simulations. The simulation environment considers the actual deployment of GWs and satellites; specifically, it is based on operational SpaceX Starlink GW sites in the United States and models the partially approved Starlink Gen2 satellite constellation by the Federal Communications Commission (FCC) [20, 21]. The orbit of the agent satellite is propagated using two-line element set (TLE) data for Starlink satellite No. 1008, retrieved on December 5, 2024. As the satellite passes over the GWs, the RSRP is periodically measured.

GW load for baseline characterization is obtained by running the distance-based FLSO algorithm over the deployed constellation snapshot and counting attached satellites at each time step. Consequently, as illustrated in Fig. 4.1, the normalized number of satellites attached per GW is visualized under random Starlink Gen2 satellite constellation placements of 60 trajectory files.

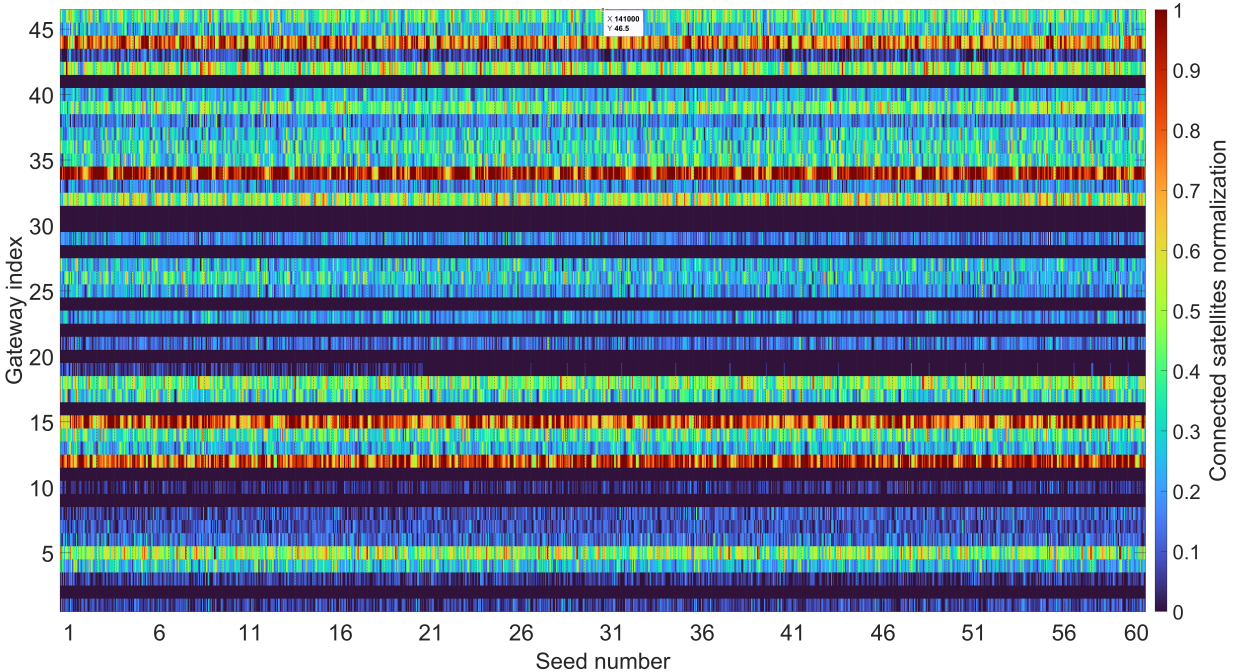


Fig. 4.1. Satellite–GW association complexity comparison (normalized satellites per GW).

The simulation parameters are summarized in Table 4.1. All algorithms share the same physical layer and mobility assumptions to ensure a fair comparison. The simulations are implemented in Python, with 1,000 independent episodes to capture the statistical behavior of LEO satellite mobility and GW association. The trained dueling DDQN requires only a few fully connected layers, resulting in sub-millisecond inference time on NVIDIA GeForce RTX 3080 GPU, which is compatible with the 200 ms measurement periodicity

The compared FLSO schemes are defined as follows.

- **Signal-based FLSO**: This scheme adapts the conventional A3 event-based HO rule to feeder link selection. A FLSO event is triggered when the RSRP of a target GW exceeds that of the serving GW by more than a configured offset. Formally, a switchover from the serving GW m_{serv} to a target GW m_{target} occurs at time t if

$$\widetilde{R}_{m_{\text{target}},n}(t) > \widetilde{R}_{m_{\text{serv}},n}(t) + \Delta R_{\text{offset}}, \tag{4.1}$$

  where $\widetilde{R}_{m,n}(t)$ is the L3-filtered RSRP from GW m to satellite n, and ΔR_{offset} is the hysteresis margin in dB. Different offsets realize different hysteresis levels. Two representative configurations are considered in the evaluation: an offset of $\Delta R_{\text{offset}} = 2$ dB, denoted as *Set 1*, and $\Delta R_{\text{offset}} = 4$ dB, denoted as *Set 2*, corresponding to moderately and strongly conservative trigger conditions, respectively. While this approach is compatible with existing new radio (NR) measurement procedures, it relies solely on instantaneous or

Table 4.1. Simulation parameters

Environment Parameter [2, 20]	**Value**
Constellation type	Walker-delta
LEO satellite altitude h	530 [km]
Inclination i	43°
Number of LEO satellites N	3360
Orbit planes P	28
Satellites per plane S	120
Phasing factor F	9
Number of GWs M	46
Carrier frequency f_m	28 [GHz]
Bandwidth B_m	480 [MHz]
GW Tx power P_m	47 [dBm]
GW Tx antenna gain G_m	49.5 [dBi]
LEO satellite Rx antenna gain G_n	38.5 [dBi]
Ambient temperature T_0	150 [K]
Antenna temperature T_a	290 [K]
Noise figure N_f	3 [dB]
Atmospheric attenuation $AT_{m,n}$	0.02 [dB/km]
Boltzmann constant k	1.379×10^{-23} [J/K]
Number of resource blocks q_{RB}	6
Minimum elevation angle $\theta_{\min}$	25°
Minimum GW RSRP $R_{\min}$	-110 [dBm]
FLSO Parameter	**Value**
Measurement period T_m	200 [ms]
MTS	1 [s]
L3 filter forgetting factor a	$(\frac{1}{2})^{K/4}$
L3 filter coefficient K	4
Distance thresholds Δd_{th}	0, 30 [km]
A3 event ΔR_{offset}	2, 4 [dB]
RL Parameter	**Value**
Learning rate	0.0001
Discount factor γ	0.99
Replay buffer limit $\mathcal{D}$	10^5
Batch size $\mathcal{Z}$	512
Total steps per episode	4,700
Polyak coefficient τ	0.01
PER exponent α	0.6
PER initial β	0.4

filtered signal strength and does not exploit geometric predictability or GW load information.

- **Distance-based FLSO**: This heuristic selects the GW with the minimum geometric distance to the satellite. A FLSO event is triggered when the distance to a target GW becomes shorter than that to the serving GW by at least a configured gap Δd_{th}. Formally, a switchover from the serving GW m_{serv} to a target GW m_{target} occurs at time t if

$$d_{m_{\text{target}},n}(t) < d_{m_{\text{serv}},n}(t) - \Delta d_{\text{th}}, \tag{4.2}$$

  where $d_{m,n}(t)$ is the Euclidean distance between GW m and satellite n at time t, and $\Delta d_{\text{th}} \geq 0$ is the hysteresis gap in km. This can be interpreted as a special case of the dual threshold distance-based framework (Events D1/D2 in [5]) with thresholds and hysteresis chosen to reduce to a simple distance-gap rule. In the evaluation, two representative configurations are considered: $\Delta d_{\text{th}} = 0$ km, denoted *Set 3*, corresponding to an aggressive closest GW selection, and $\Delta d_{\text{th}} = 30$ km, denoted *Set 4*, enforcing a more conservative gap to suppress frequent switchovers. This provides a standard aligned yet purely geometry-driven feeder link selection baseline.

- **DQN-based FLSO**: The DQN baseline is trained using the same state, action, and reward definitions as the proposed dueling DQN agent, but with a conventional single stream Q-network. Learning

uses TD updates with experience replay and a target network. After training, the learned Q-network is frozen and evaluated without further exploration.

4.2. Key Performance Indicators

This thesis evaluates five key performance indicators (KPIs) that together capture FLSO stability, link quality, and GW load balancing. Unless otherwise noted, all metrics are computed per episode and averaged over time and satellites.

FLSO measures how many times, on average, each satellite changes its serving GW during an episode. It reflects the overall switching activity of the FLSO policy; smaller values indicate a more stable association with fewer disruptions and reduced control overhead.

UFLSO refines this by counting switchovers that are followed by a very short dwell time, shorter than a predefined MTS. Such events correspond to ping-pong or impulsive decisions that do not yield lasting performance gains. This KPI quantifies how many switches are effectively wasteful; a desirable policy minimizes both the total number of switchovers and the fraction that are unnecessary.

ToS denotes how long a satellite continuously remains connected to the same GW between two consecutive switchovers. This thesis reports the average dwell time over all satellites and all stays. Larger ToS values indicate that the policy maintains more persistent feeder link associations, reducing signaling overhead and limiting service interruptions

while still allowing beneficial HOs.

RSRP evaluates the long term quality of the selected feeder links. At each time step, this thesis observes the L3 filtered RSRP of the GW actually serving each satellite, using the standardized IIR-based parameters in Table 4.1. This thesis then averages this feeder link RSRP across all satellites and time. Higher values mean that satellites are more consistently attached to GWs providing strong received power, showing that stability is not obtained at the cost of degraded link quality.

GW load characterizes how evenly the feeder link load is distributed across GWs. For each GW, this thesis monitor how many satellites are attached over time, summarize this as its average load, and track the maximum load observed across all GWs and time. Lower peak load and a tighter spread of average loads indicate more balanced utilization of feeder link resources and a reduced risk of persistent GW bottlenecks.

4.3. Results and Discussion

Fig. 4.2 illustrates the normalized episodic reward during training for the proposed FLSO agent and the DQN baseline. Both agents quickly surpass a normalized reward of 0.9, confirming that they learn feasible FLSO policies; however, the proposed agent converges faster, attains a higher steady state reward, and exhibits clearly reduced variance without the deep reward drops observed for DQN. This behavior is consistent with its architecture: the dueling structure stabilizes value estimation by separating state value and action advantage, the DDQN update mit-

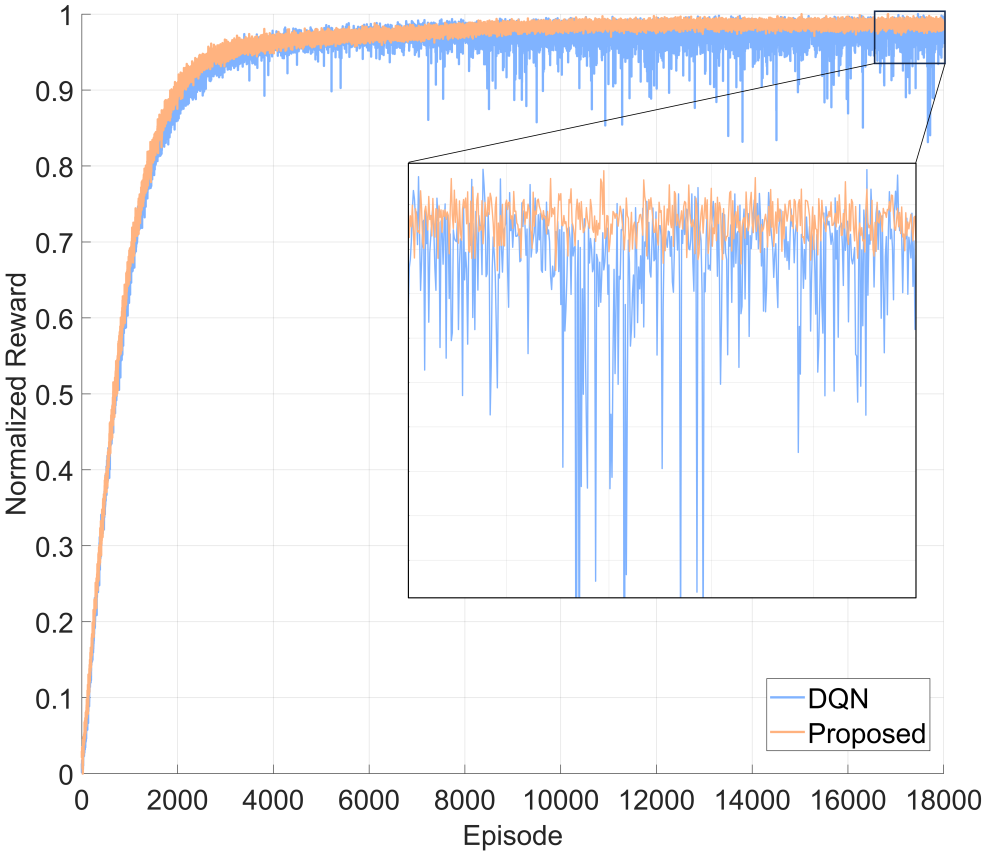


Fig. 4.2. Average reward over episodes for the proposed and DQN-based FLSO.

igates overestimation induced oscillations, and prioritized replay with Polyak averaged targets smooths learning against non-stationary LEO geometry and GW load. In contrast, the baseline DQN, relying on max based targets and uniform replay, is more prone to transiently overvaluing aggressive switching or overloaded GWs, which translates into bursts of UFLSOs, shortened ToS, and load imbalancing actions that appear as intermittent reward collapses. Thus, the reward evolution directly reflects

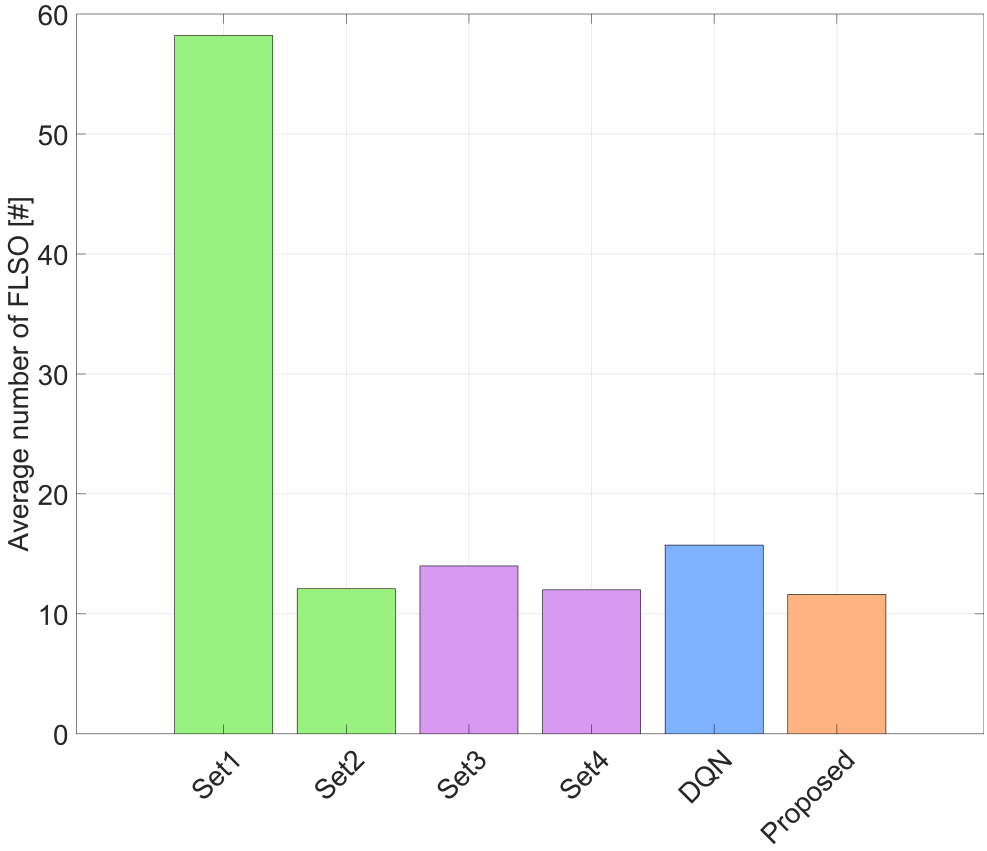


Fig. 4.3. Average number of FLSO for the proposed, DQN-based, distance-based, and signal-based FLSO.

that the proposed agent internalizes the multi objective design stability, link quality, and GW load awareness more reliably than the conventional DQN.

Fig. 4.3 provides a detailed comparison of the average number of FLSOs across the considered schemes. The signal-based configuration with a low offset, denoted *Set 1*, with a 2 dB offset, results in extremely aggressive behavior, with an average of 58.21 FLSOs per episode. In-

creasing the offset to 4 dB in *Set 2* dramatically reduces the FLSO count to 12.09, confirming the strong sensitivity of A3 type rules to hysteresis tuning. The distance-based baselines show a similar pattern: the closest GW rule in *Set 3*, where $\Delta d_{\text{th}} = 0$ km, yields 14 switchovers on average, whereas introducing a 30 km gap in *Set 4* lowers this to 12, at the price of relying on a hand-crafted geometric margin. The plain DQN agent converges to 15.73 switchovers and thus remains more reactive and unstable than all well tuned heuristic sets. In contrast, the proposed dueling DDQN-based FLSO achieves the lowest FLSO frequency at 11.62 without depending on manually chosen offsets or distance gaps. These results indicate that the proposed RL design internalizes an effective stability performance trade-off, matching or outperforming carefully tuned heuristics and clearly improving upon the baseline DQN.

Fig. 4.4 evaluates the number of UFLSOs. The low offset signal-based configuration, *Set 1*, is clearly unstable, yielding on average 4.56 UFLSOs per episode, that is, multiple switchovers that immediately violate the MTS constraint and represent pure ping-pong behavior. Increasing the A3 offset in *Set 2* or introducing distance gaps in the geometry based rules, such as *Set 3* and *Set 4*, completely suppresses UFLSO events, but at the cost of more conservative decision policies. For the distance-based sets, this is structural: once a FLSO is triggered because a target GW is closer than the serving GW by at least Δd_{th}, the relative geometry under LEO motion does not revert in favor of the previous serving GW within the short MTS window, so rapid back switching cannot occur and UFLSO is eliminated by design. The DQN baseline still

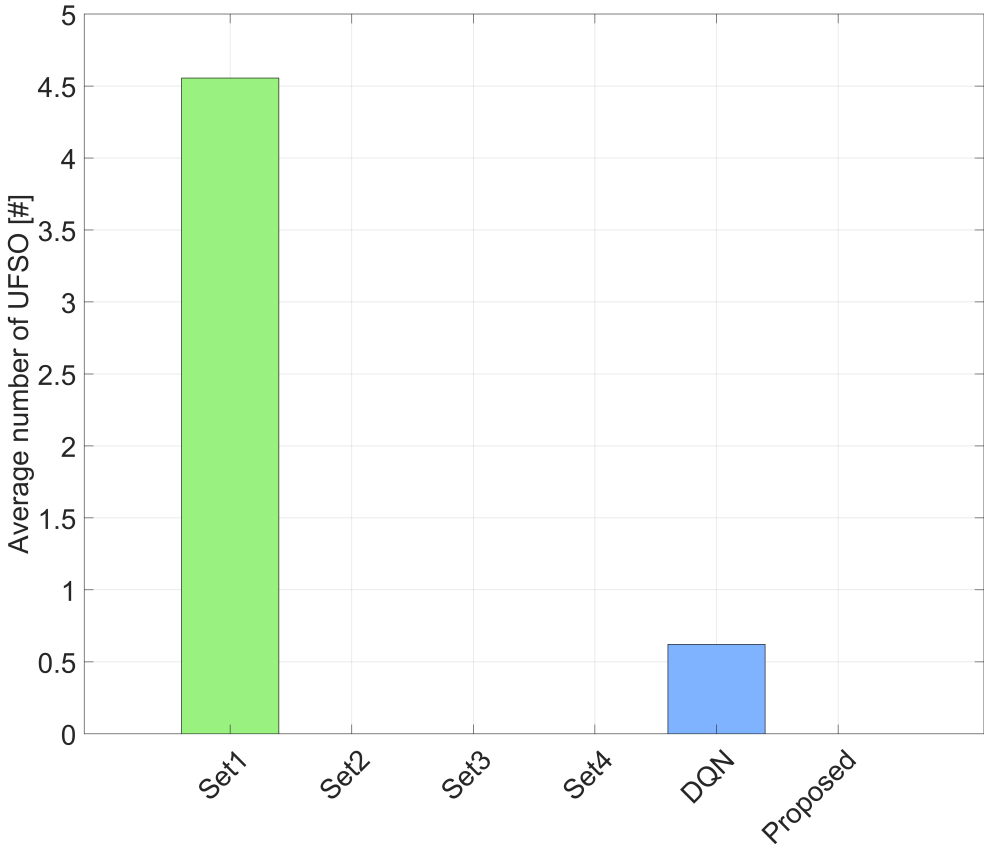


Fig. 4.4. Average number of UFLSO for the proposed, DQN-based, distance-based, and signal-based FLSO.

exhibits an average of 0.62 UFLSOs per episode, indicating residual oscillatory decisions despite learning. The proposed dueling DDQN FLSO achieves *zero* UFLSO while also maintaining the lowest FLSO level, confirming that it learns to perform only persistent and beneficial switchovers rather than relying on ad-hoc hysteresis tuning.

Fig. 4.5 compares the average ToS across all schemes. The aggressively reactive signal-based *Set 1* yields the shortest ToS of 928.36 s, more

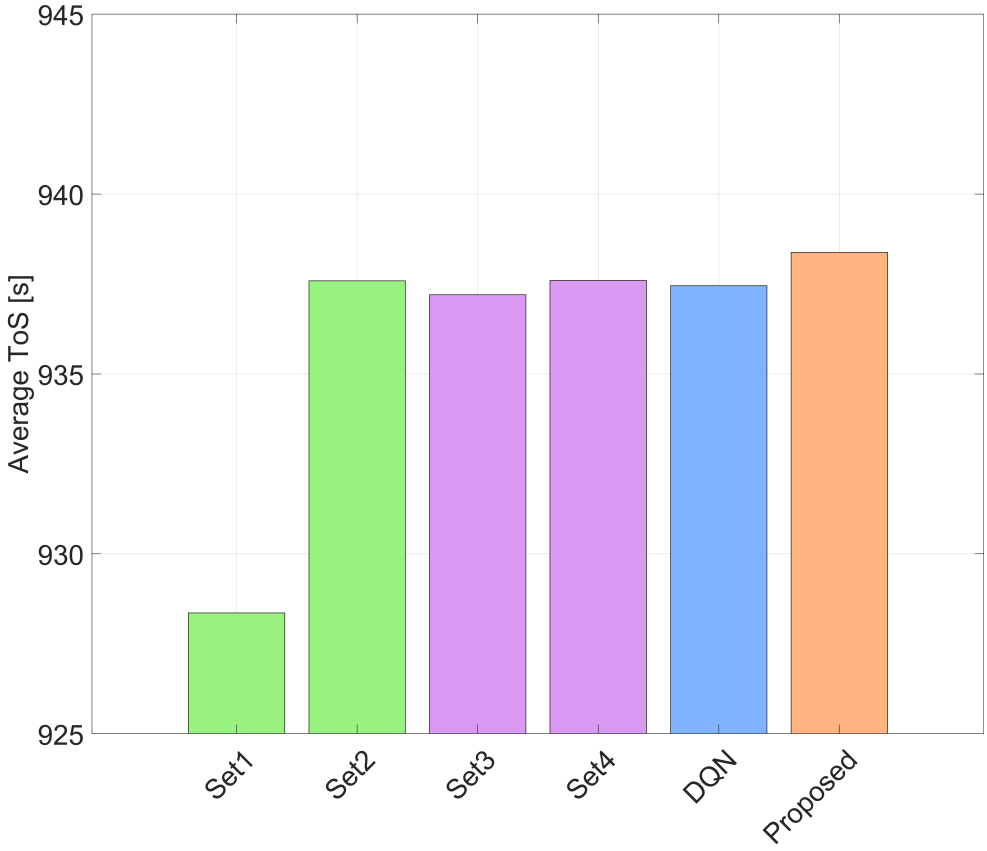


Fig. 4.5. Average ToS for the proposed, DQN-based, distance-based, and signal-based FLSO.

than 10 s lower than the other configurations and consistent with its excessive FLSO and UFLSO. Increasing the offset in *Set 2* raises the ToS to 937.59 s, which is comparable to the distance-based *Set 3* with 937.20 s, *Set 4* with 937.60 s, and the DQN baseline with 937.45 s; once ping-pong behavior is mitigated, these schemes all maintain long associations close to the episode duration. The proposed algorithm further improves the ToS to 938.38 s, the longest among all candidates, while simultaneously

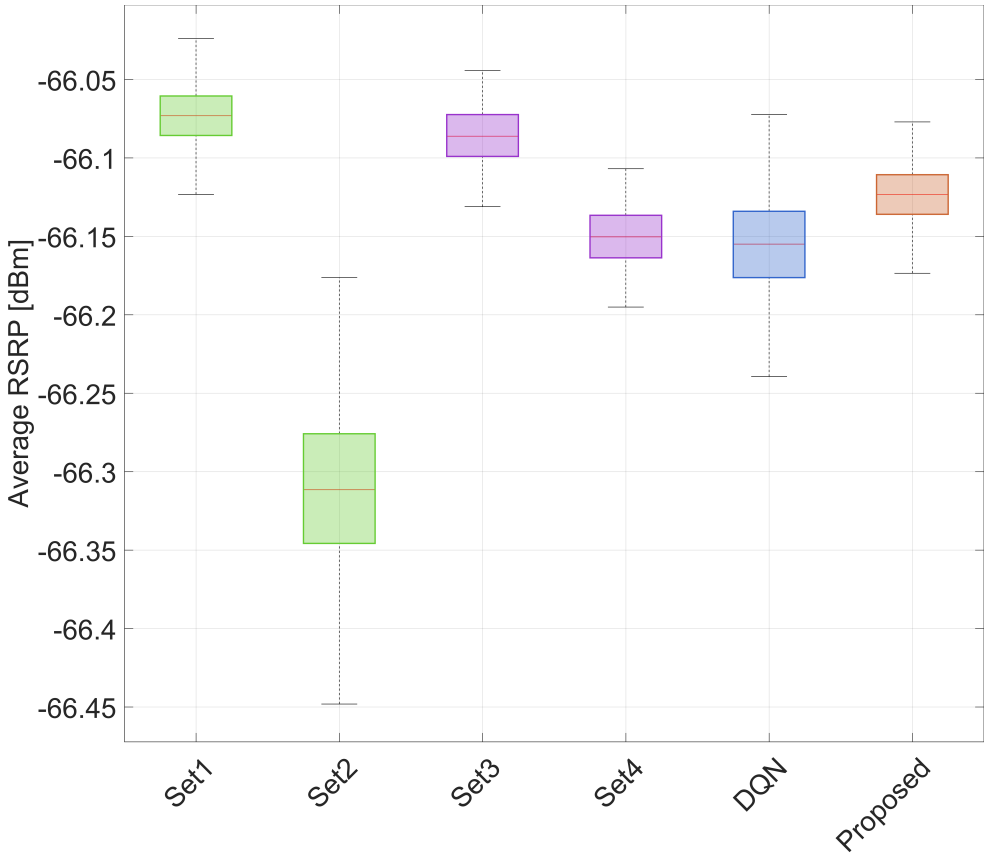


Fig. 4.6. Average RSRP for the proposed, DQN-based, distance-based, and signal-based FLSO.

achieving the lowest FLSO and zero UFLSO. This demonstrates that the dueling DDQN policy effectively suppresses superfluous switchovers and maximizes the persistence of beneficial links, enabling stable feeder link operation without relying on overly conservative hysteresis.

Fig. 4.6 illustrates the feeder link RSRP performance based on the summary statistics. The signal-based configuration *Set 1* with a 2 dB offset achieves the highest average RSRP of approximately -66.07 dBm

with very small variance; however, as shown in Figs. 4.3 and 4.4, this gain is obtained at the cost of extremely frequent and unnecessary FLSOs and UFLSOs and is therefore impractical. Increasing the offset in *Set 2* to 4 dB yields the worst link quality among all schemes, with the mean RSRP dropping to about -66.31 dBm and the dispersion becoming larger, which confirms that an overly strong hysteresis forces satellites to remain attached to outdated GWs. The distance-based *Set 3* and *Set 4* achieve average RSRP values of approximately -66.09 dBm and -66.15 dBm, respectively, with tight spreads, indicating that geometry driven selection can maintain reasonable link quality when the distance threshold Δd_{th} is appropriately configured, although GW load remains unaccounted for. The DQN baseline attains a mean RSRP of about -66.16 dBm with a wider lower tail, which is consistent with the reward drops and residual policy instability observed during training. The proposed dueling DDQN scheme achieves an average RSRP of approximately -66.12 dBm, located between the distance-based configurations and superior to *Set 2* and the DQN baseline, with a compact distribution and no severe outliers, while simultaneously realizing the lowest FLSO count and zero UFLSO. These results demonstrate that the proposed policy sustains near optimal link quality relative to practical baselines, avoids high RSRP yet high switching behavior, and operates at a more favorable trade-off between feeder link power, switching overhead, and stability.

Fig. 4.7 highlights how each scheme distributes satellites across GWs. The proposed method yields the lowest mean load of 1.99 satel-

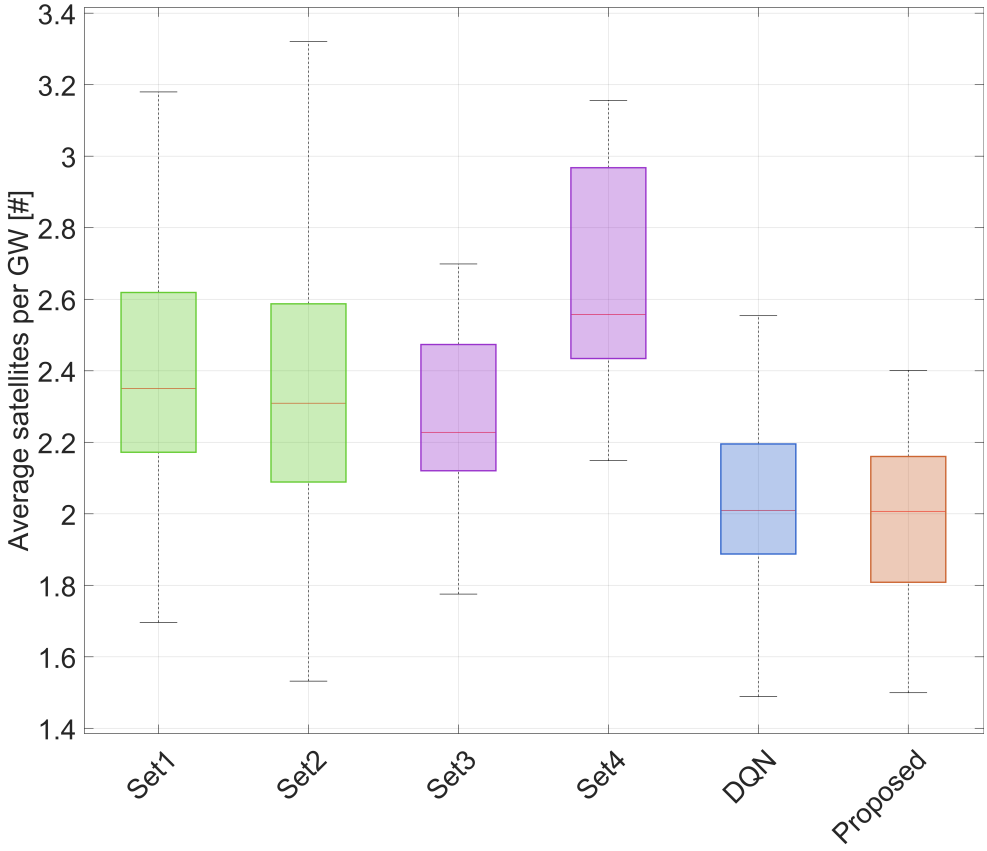


Fig. 4.7. Average satellites per GW for the proposed, DQN-based, distance-based, and signal-based FLSO.

lites per GW and the smallest standard deviation of 0.24, indicating that almost all GWs operate close to a common load level and that persistent hot spots are effectively avoided. The DQN baseline is the second most balanced, with a mean of 2.03 and a standard deviation of 0.26, but it still exhibits a wider spread, which implies residual irregularity in its association policy. The distance-based heuristics are more skewed: *Set 3* has a mean of 2.28 and a standard deviation of 0.27 and already con-

centrates more satellites on specific GWs, while *Set 4*, configured with a larger Δd_{th}, increases both the mean to 2.66 and the standard deviation to 0.33, revealing that sticky geometric preferences amplify spatial imbalance. The signal-based schemes show a similar tendency: *Set 1* yields a mean of 2.38 and a standard deviation of 0.31, resulting in volatile and uneven loads consistent with its excessive FLSOs, whereas *Set 2* reduces the mean to 2.35 but inflates the standard deviation to 0.35, which is the largest variance among all practical baselines. Together with the FLSO, UFLSO, and RSRP results, these findings confirm that the proposed dueling DDQN policy provides the most uniform and efficient utilization of GW resources while maintaining stable and high quality feeder link associations.

Fig. 4.8 further characterizes the GW load distribution by plotting the empirical CDF of the time averaged number of attached satellites per GW. The curve for the proposed scheme is the steepest and is tightly concentrated around two satellites per GW, indicating that the vast majority of gateways experience very similar loads and that heavy load outliers are effectively eliminated. The DQN baseline shows a similar but slightly right-shifted and flatter profile, revealing residual irregularity in its GW association decisions. In contrast, the signal-based configurations, in particular Set 1, exhibit a broader CDF with noticeable probability mass at higher load levels, reflecting volatile and uneven load patterns induced by excessive switching. The distance-based schemes, especially Set 4 with a large Δd_{th}, are shifted further to the right and display the longest upper tails, confirming that sticky geometry driven associations cause a subset

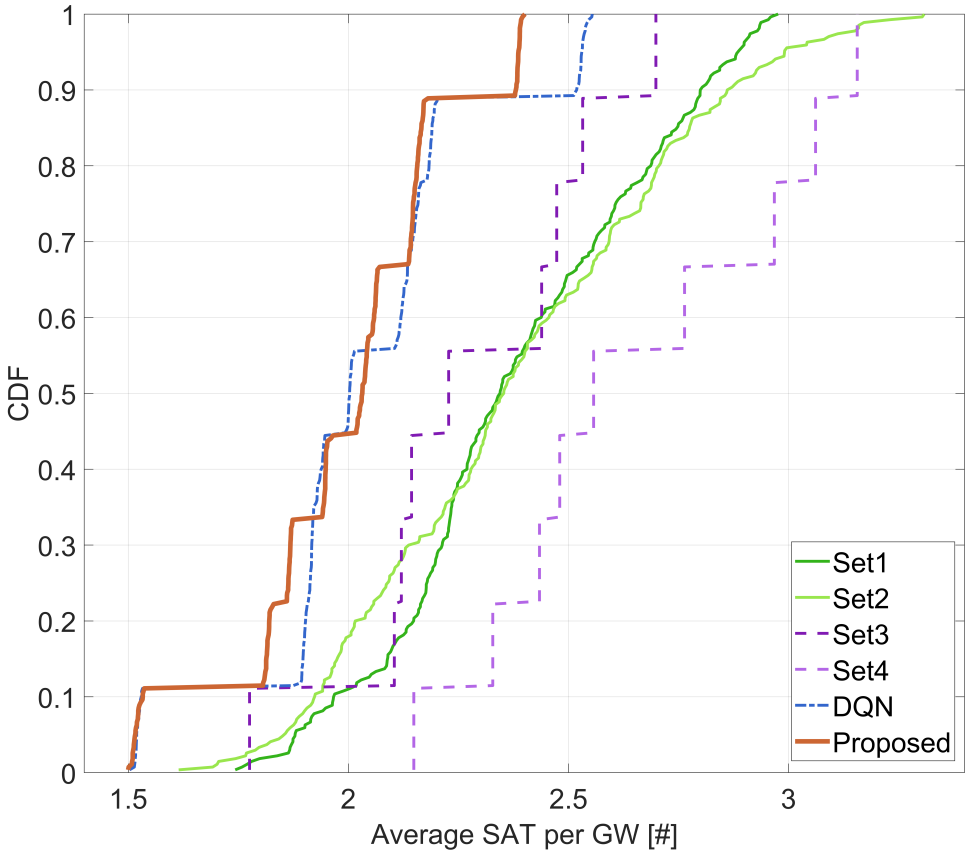


Fig. 4.8. CDF of average satellites per GW for the proposed, DQN-based, distance-based, and signal-based FLSO.

of GWs to persistently attract more satellites. Together with the summary statistics in Fig. 4.7, this CDF based view corroborates that the proposed dueling DDQN policy not only minimizes switching overhead but also most effectively equalizes the feeder link load across GWs.

Chapter 5. Conclusion

This thesis has investigated RL-based FLSO for GW load balancing in LEO NTN, focusing on regenerative payload architectures where backhaul mobility is decoupled. Building on practical assumptions including a Walker-delta constellation emulating the Starlink Gen2 deployment, realistic GW locations in the United States, and 3GPP-compliant measurement and filtering procedures—the proposed framework targets the non-standardized but performance-critical problem of selecting the serving GW under rapidly time-varying LEO geometry.

The FLSO decision problem was formulated as a constrained MDP that jointly accounts for link stability, UFLSOs, feeder link quality, and GW load imbalance. To solve this problem, this thesis designed a dueling DDQN-based FLSO policy with PER and Polyak-averaged target updates. The state incorporates ephemeris-aware geometry, L3-filtered RSRP, previous association, and instantaneous GW load, while the reward encodes a multi-objective trade-off across FLSO and UFLSO penalties, ToS, link quality, and load regularization. Conceptually, the proposed method extends 3GPP distance-based triggering, such as D1/D2-type events, toward a GW-aware, learning-based FLSO policy that re-

mains compatible with standardized measurement abstractions but is no longer limited to purely link-centric heuristics.

Extensive simulations confirm that the proposed dueling DDQN policy consistently outperforms both heuristic and standard DQN baselines across all evaluated key performance indicators. Compared with the signal-based FLSO, the proposed scheme avoids the counterproductive scenario where high RSRP measurements lead to excessive switching behavior: it eliminates UFLSOs, suppresses excessive FLSOs without relying on overly conservative offsets, and sustains near-optimal RSRP. Relative to distance-based schemes interpreted the proposed policy achieves comparable or better link quality and longer ToS while significantly improving GW load balancing, avoiding the spatial skewness and hotspot formation induced by purely geometric selection. Against the DQN baseline trained under the same environment, the proposed architecture converges faster, exhibits fewer reward collapses, yields lower FLSO and UFLSO, and achieves more uniform GW utilization, demonstrating the effectiveness of the dueling DDQN design in stabilizing learning under non-stationary LEO dynamics.

Overall, the results show that a lightweight, distributed RL-based FLSO controller can operate within realistic measurement periodicities and leverage locally observable metrics to realize a more favorable trade-off among switching stability, backhaul link quality, and GW load balancing than both standardized-style heuristics and conventional value-based agents.

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